

$$\begin{aligned}
&\equiv (\forall x)(\forall y)((\forall z) \neg (P(x,z) \wedge P(y,z)) \vee (\exists u) Q(x,y,u)) \quad (2) \\
&\equiv (\forall x)(\forall y)((\forall z)(\neg P(x,z) \vee \neg P(y,z)) \vee (\exists u) Q(x,y,u)) \quad (\text{1st De Morgan's Law}) \\
&\equiv (\forall x)(\forall y)((\forall z)(\neg P(x,z) \vee \neg P(y,z) \vee (\exists u) Q(x,y,u))) \quad (3) \\
&\equiv (\forall x)(\forall y)((\forall z)(\exists u) Q(x,y,u) \vee (\neg P(x,z) \vee \neg P(y,z))) \quad (\text{1st Commutative Law}) \\
&\equiv (\forall x)(\forall y)(\forall z)(\exists u)(Q(x,y,u) \vee (\neg P(x,z) \vee \neg P(y,z))) \quad (3)
\end{aligned}$$